# Parth Desai

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## TECHNICAL SKILLS

**Languages :** C++, Python, MATLAB, Dart, SQL
**Robotics & Simulation :** ROS2, RViz, Gazebo, MuJoCo, Simulink, Simscape
**ML & CV Libraries :** NumPy, Pandas, Keras, Tensorflow, OpenCV, Matplotlib, Seaborn, Scikit-Learn, Tkinter
**Frameworks & Tools:** Flutter, Github, Github, Gitlab, Firebase

## EXPERIENCE

**Software & Automation Developer** | *Rashmi Enterprises* | *Jan 2021 - Dec 2021, July 2022 - July 2023* ***India***

- Designed and deployed the company's website, establishing a professional digital presence and improving **client outreach by 50%**.
- Developed an **OpenCV**-based **print defect detection system** prototype, identifying **misalignments and smudges** in **real-time** using the integrated camera footage.
- Simulated a **fleet of autonomous mobile robots (AMRs)** in **ROS2** and **Gazebo** to **automate material transport** across print, binding, and packaging stations using **decentralized task coordination**.
- Developed a **vision-guided sorting prototype**, where agents classified printed materials by quality using **OpenCV** and executed sorting tasks through **ROS2-controlled actuators**.
- Designed a **modular ROS2 architecture** with separate **perception**, **planning**, and **control nodes**, enabling scalable and testable simulation workflows.

**Flutter Development Intern** | *AmazeYoo / Arkverse Pvt. Ltd.* | *Jan. 2022 - June 2022* ***India***

- Revamped **UI/UX** with **reusable Flutter custom widgets**, increasing **user engagement by 90%.**
- Designed and implemented **RESTful APIs** using **Django REST Framework** and **PostgreSQL** for efficient backend integration.
- Developed a **Python-based recommendation engine** achieving **93% accuracy** in content personalization.

## PROJECTS

**HELIOS SAR Drone *(Ongoing)*** | *ROS2, Python, Gazebo, Linux*

- Developing a search and rescue drone capable of **autonomous navigation** in disaster environments.
- Implemented **SLAM** for real-time mapping of custom disaster-themed Gazebo worlds.
- Engineered **path planning using RRT*** for obstacle avoidance using **multi-modal sensor input** (**LIDAR, Camera, IMU**).
- Designing **computer vision pipelines** for **object detection** and **victim localization** using **OpenCV** and **ROS2**.

**Real Steel** | *Python, MuJoCo, Linux*

- Built a **camera-based motion capture system** using **MediaPipe** for robust **3D pose estimation** with **orientation correction**.
- Developed an **analytical inverse kinematics module** to convert human poses into robot joint angles under kinematic constraints.
- Integrated a modular real-time **control pipeline** integrating **perception, kinematics, and actuation layers** in MuJoCo for responsive humanoid behavior.
- Implemented **motion smoothing** and **angle correction** for stable and natural robot behavior during dynamic tasks.

**Generative Enhanced Noise Cancellation & Signal Improvement System** | *Python*

- **Improved speech clarity in noisy environments** to support users with hearing impairments and auditory autism.
- Designed and implemented the GENESIS architecture with **generator and discriminator networks**.
- Designed a **GAN-based system** using **encoder/decoder** architecture with **CNNs and dilated DenseNet.**
- Applied **MetricGAN** for **objective speech quality enhancement** using self-recorded, real-world noisy audio samples.
- Tested system in environments like cafes, traffic, and music-filled spaces to simulate real use cases.

## EDUCATION

**Iowa State University** Ames, IA
*MS in Computer Science* **(CGPA: 3.78/4)** *Aug 2023 - Aug 2025*
**Relevant Coursework:**
*Motion Planning, Computational Perception, Principles of AI, Machine Learning*